TypeScript Assignments – Objects

Object Literals

1. Create a student object with name, age, and course properties.
2. Create a car object with brand, model, and price.
3. Create an employee object containing id, name, department, and salary.
4. Create a product object containing id, name, price, and inStock.
5. Create a bank account object containing accountNumber, holderName, and balance, and print all properties.

Object Types

6. Define a type for a student object and create one student using that type.
7. Define a type for a product with id, name, and price.
8. Define a type for an employee containing id, name, department, and salary.
9. Create a type for a book containing title, author, price, and isAvailable.
10. Create a type for a bank account and write a function that accepts the object and prints the account details.

Destructuring

11. Create a student object and extract name and age using object destructuring.
12. Create a product object and extract name and price.
13. Create an employee object and destructure three properties from it.
14. Create a nested object containing address information and extract the city using destructuring.
15. Create an object with five properties and use destructuring to extract only three of them.

Spread Operator

16. Create two objects containing student information and combine them using the spread operator.
17. Create a student object and create a new object with the same properties plus course.
18. Create two arrays of numbers and combine them using the spread operator.
19. Create a product object and create a copy with a different price using the spread operator.
20. Create a user object and create a new object where only the role property is changed.

Rest Operator

21. Create an object with name, age, city, and course. Extract name and store the remaining properties using rest.

22. Create an object containing employee information and extract id, keeping the remaining properties in another object.
23. Create an array of numbers and use rest to separate the first number from the remaining numbers.
24. Create an array of student names and use rest to separate the first two students from the remaining students.
25. Create an object with at least five properties and use destructuring with rest to create an object containing all properties except two.

Readonly Properties

26. Create a student type where id is readonly. Try to modify the id.
27. Create a product type where productId is readonly but price can be changed.
28. Create an employee object where employeeId cannot be modified.
29. Create a User type with readonly username and mutable email. Modify both and observe the result.
30. Create a bank account type where accountNumber is readonly but balance can be modified.

Optional Properties

31. Create a Student type where email is optional.
32. Create a User type with required name and optional phone.
33. Create a product type where discount is optional. Create products both with and without discount.
34. Create an employee type where department is optional.
35. Create a function that accepts a user object with an optional phone property and prints the phone only when it exists.